

When the Model Screens Off the Harness: Per-Decision Cross-Layer Causal Credit in LLM Agents

[TODO: autores]

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Abstract

LLM agents interleave decisions from two layers: the *model* (tool calls) and the *harness* (context management, retries, termination). We show, via deterministic counterfactual replay with a *measured* noise floor (exact fidelity under explicit serving premises), that per-decision causal contributions $C(H)$, $C(M)$ and their interaction $I(H, M) = C(H, M) - C(H) - C(M)$ are identifiable on the same trajectory. Interaction is *regime-dependent*: with budget slack, the model action screens off the harness decision ($C_{HM} = C_M$, 0 breaks in 57 points across three configurations); under budget pressure, screening-off breaks (8/65 across three pressure levels; $p = 0.005$ point-level, $p = 0.125$ task-clustered — we report both) at *recurring* decision points, with a non-monotonic dependence on pressure intensity partially mediated by reward saturation. The noise floor and the slack regime replicate on a second model (Qwen3-1.7B); the pressure regime is established on one model — the second model’s null is itself explained by saturation (1 non-saturated point). This implies that single-layer credit double-counts shielded decisions. We then train the harness layer with interaction-aware credit against outcome-only, single-layer, and untrained controls (dose-matched in total rollouts, 3 seeds, twice — the second run with a pre-registered optimization fix): no credit signal learns the threshold policy that is expressible by the policy class and dominant on held-out tasks (the threshold policy reaches held-out R_{eff} 0.398; every trained arm converges to keep-always at 0.237 or collapses), and credit-driven arms add collapse risk that the outcome-only arm avoids. A pre-registered audit (60 points, two-greedy-replay recomputation, 58/60 sign agreement) rules out the estimator as the cause: the binding constraint is *state coverage of the training task distribution*, not credit quality — no context policy reaches the states where the threshold policy pays on the training tasks. We report this two-stage identified negative result as a design lesson for cross-layer training benchmarks.

1 Introduction

An LLM agent is not a model. It is a model wrapped in a *harness*: a program that manages context, decides when to summarize or truncate history, retries failed calls, and terminates episodes. In deployed agents the harness makes as many decisions per episode as the model does, and those decisions — especially context management — destroy or preserve the information the model conditions on. Yet the dominant training paradigm treats the trajectory as a monolith: outcome-level RL (e.g., GRPO-style policy gradients) assigns one scalar return to every decision, regardless of which layer made it, and recent per-step credit methods attribute within a single layer only. If a harness decision was harmless *because* the model’s next action compensated for it, single-layer credit will bill the harness for an effect the model screened off — a double-counting error that is invisible unless both layers are measured on the same trajectory.

This paper asks the question directly: on the same trajectory, what is the causal contribution of a harness decision, of a model decision, and of their interaction? We answer it with deterministic counterfactual replay. Given a recorded trajectory, we re-execute it from any decision point with exactly one decision changed and the entire prefix forced, and define per-decision credit as $C(d) = R_{\text{orig}} - R_{\text{cf}}$. Replaying pairs and joint flips yields $C(H)$, $C(M)$, $C(H, M)$ and the interaction $I(H, M) = C(H, M) - C(H) - C(M)$ at a single decision pair. Because our replay is *exactly* deterministic (a measured, not assumed, property — and one that fails silently if vLLM’s automatic prefix caching is left on), every nonzero credit is signal.

The headline finding is that the interaction term is *regime-dependent*. When the agent has budget slack, the model action screens off the harness decision exactly: $C_{HM} = C_M$ in 57 out of 57 measured points across three configurations. Under budget pressure the shield cracks — 8 breaks in 65 points, concentrated at recurring decision points — and the dependence on pressure intensity is non-monotonic, partially mediated by reward saturation: interaction is only detectable inside a *competence window* where the agent neither aces nor fails everything. The practical corollary is negative for naive pipelines: in the slack regime, single-layer harness credit systematically double-counts effects the model absorbs.

We then test whether interaction-aware credit improves *training* of the harness layer, with pre-registered predictions, dose-matched controls, and three seeds per arm — twice. The honest outcome is a two-stage identified negative: optimization instability first, and, after a pre-registered fix, stable training that still fails to discover a threshold policy which is expressible by the policy class and dominant on held-out tasks. A pre-registered audit exonerates the credit estimator; the binding constraint is state coverage, not attribution. We report this as a design lesson for cross-layer training benchmarks, not as a method contribution.

Contributions.

1. **Deterministic replay infrastructure with a measured noise floor** under explicit identification premises (fixed config, sequential execution, prefix caching off) — including a cautionary finding: vLLM automatic prefix caching silently breaks greedy-decoding determinism across differing serving histories.
2. **Per-decision counterfactual decomposition** of $C(H)$, $C(M)$, and $I(H, M)$ on the same trajectory, and the finding that screening-off is regime-dependent (exact under slack, breaking at recurring points under pressure).
3. **Anatomy of screening-off**: a structural component (the sampled a' re-injects destroyed information — a property of the estimand, 23/24 annotated points) and an empirical component (the live downstream harness re-triggers the intervened decision, 22/52 points).
4. **Construct validation by control tasks**: the method separates harmless from harmful harness decisions by construction (11/12), including one audited exception where a harness decision had a *behavioral*, not informational, effect.
5. **A pre-registered training comparison with an identified negative outcome**: interaction-aware credit vs. outcome-only, single-layer credit (a reproduction of the CHILL-Harness principle in our environment), and a zero-credit masking control (de facto an untrained keep-always policy under our greedy tie-break — stated explicitly), dose-matched in total rollouts. Act 1: every trained arm collapses seed-dependently. Act 2 (after a pre-registered optimization fix): stable training, right direction, yet no arm discovers the threshold policy that is expressible and dominant on held-out tasks, and credit-driven arms retain collapse risk (1/3 seeds) that outcome-only avoids (0/3). A pre-registered audit (58/60 sign agreement) rules out the estimator; state-visitation analysis localizes the failure in training-distribution state coverage, not exploration pressure.

2 Related Work

Counterfactual replay for agent credit is an active area, but every existing system we are aware of attributes within a single layer. CHILL-Harness [3] intervenes on harness decisions and allocates replay budget adaptively, but never measures the model layer or the interaction; our single-layer training arm (§6) is an explicit reproduction of its principle in our environment, which turns the comparison into a result. CAR [2] formalizes per-step do-operations with Shapley attribution under a replay budget, within one decision stream. C3 [1] audits credit fidelity against exact replay ground truth (the same validation metric we adopt) but decomposes across *agents*, not across the model/harness boundary inside one agent. A recent negative result [7] shows that popular step-level credit signals fail to beat chance against replay ground truth; it sets the evidential bar

Method	counterfactual via replay	measures $C(H)$	measures $C(M)$	$I(H, M)$ on same trajectory	trains with I
CHILL-Harness	✓	✓	—	—	—
CAR	✓	—	✓	—	—
C3 v2	✓	(per agent)		—	—
HASE	—	✓	—	—	—
Ours	✓	✓	✓	✓	tested [†]

Table 1: T1 — positioning. [†]Tested with dose-matched controls and pre-registration; the outcome is an identified negative (§6). [TODO: verify cells against primary sources before submission]

our critic section inherits (dose-matching, pre-registration, heuristic baselines). On the training side, Co-Harness [4], HarnessCompass [5] and HASE [6] jointly optimize harness and weights, but without a per-decision causal signal — they optimize the composition blindly. Agent Lightning [8] provides rollout infrastructure for agent RL without addressing attribution. Table 1 summarizes: no prior work measures $C(H)$, $C(M)$ and $I(H, M)$ on the same trajectory, which is precisely where the double-counting error lives.

3 Setup: A Two-Layer Agent with Deterministic Replay

3.1 Agent, decision schema and recorder

The agent is deliberately minimal: a coding agent in a sandboxed environment that reads a task description, writes files, and runs tests. Two layers make decisions. The **model layer** (Qwen3-4B served by vLLM, greedy decoding) chooses tool calls: `write_file`, `run_tests`, `read_file`, `finish`. The **harness layer** is a scripted policy that at every turn makes a `context_policy` decision (`keep_context` vs. `summarize_context`, the latter compressing history to a fixed-length prefix) and a `termination` decision. Every decision — model or harness — is recorded with a typed schema: decision id, layer, state before, available actions, chosen action, observation, token costs, and parent/child links. Reward is deterministic: the fraction of hidden pytest tests passed, so $R \in [0, 1]$ with resolution set by the number of tests. Cost-adjusted reward is $R_{\text{eff}} = R - \lambda \cdot \text{prompt tokens}/10^5$ with $\lambda = 25$, fixed analytically before any training and never recalibrated.

3.2 Replay and interventions

Replay re-executes a recorded trajectory through the *live* agent loop with a forced-action queue: recorded decisions are matched against the loop’s requests (peek-match on decision type and state hash) and forced, up to the intervention

point; from there the loop runs free. An intervention replaces exactly one decision (or, for $C(H, M)$, one harness-model pair). The null intervention — force everything, change nothing — must reproduce R exactly; §4.1 shows it does, and what serving configuration is required for it to. All replays run sequentially against a single vLLM instance with a fixed config.

3.3 Tasks

We use 30 synthetic coding tasks in four designed strata, plus 10 held-out tasks. **V2** and **L** tasks place critical constants beyond the first 240 characters of history, so a summarize decision is *informationally destructive* at known positions; **L** tasks are longer variants. **S** tasks are built so that success requires harness and model decisions to cooperate (synergy by design). **C** tasks are controls: all information needed survives any summary, so the construct predicts $C(H) = 0$. Reward saturation (episodes at $R \in \{0, 1\}$) is endogenous to the number of tests per task; we treat it as a design choice and analyze its consequences explicitly (§4.3, §7).

4 Measuring Cross-Layer Credit

4.1 Noise floor

Every causal claim below is only as good as the replay. We therefore *measure* the noise floor rather than assume it: the pre-registered floor for each configuration is $\max |\Delta R|$ over null replays (replay with no intervention). The floor is defined on the reward R , not on trace equality: in 3 null replays (all in one configuration) the replayed trajectory took one extra retry while reproducing R exactly, and we report this rather than hide it. Across seven dedicated null-replay runs of Qwen3-4B (1,230 nulls), the nulls attached to every interaction run (401 more on Qwen3-4B), and two configurations of Qwen3-1.7B (64 nulls), **every one of 1,695 null replays reproduced R exactly**: the floor is 0.0 in all configurations (Figure 1a). This is a measurement under explicit premises — fixed serving config, strictly sequential requests, prefix caching off — not a guarantee.

The premises matter. vLLM V1 enables automatic prefix caching (APC) by default, and with APC on, prefill numerics depend on the KV-cache state left by *previous* requests: greedy decoding diverges across identical prompts with different serving histories, even when requests are sequential. In our APC-contaminated run, 7 of 270 null replays were inexact and the floor rose to 0.417 (Figure 1b) — large enough to swamp most true credits. Sequential execution alone is *not* sufficient; APC must be disabled. We flag this because any replay-based attribution system built on default vLLM settings inherits this silent failure.

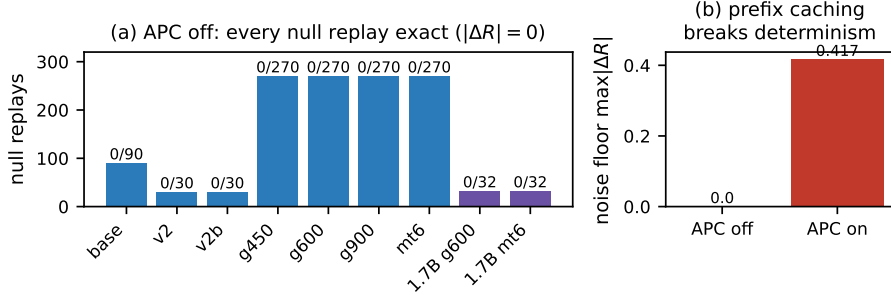


Figure 1: **F1 — Measured noise floor.** (a) Dedicated null-replay runs (Qwen3-4B blue, Qwen3-1.7B purple); together with nulls attached to interaction runs, all 1,695 null replays are exact, floor = 0.0. (b) With vLLM automatic prefix caching on, the floor rises to 0.417 (7/270 inexact): greedy determinism breaks silently.

4.2 $C(H)$ and $C(M)$

With an exact floor, per-decision credits are directly interpretable: any $C \neq 0$ is a real causal effect of that single decision. Figure 2 shows the distributions over 248 harness points and 66 model points, by stratum. Three regularities matter downstream. First, direction is everything: flipping **keep**→**summarize** at an information-critical point yields large positive $C(H)$ (the decision to keep was worth up to +0.89), while the reverse direction is near zero — we therefore never aggregate across directions. Second, credits are sparse: most decisions are causally inert, and mass concentrates at information-destruction points predicted by task design (V2/L strata). Third, $C(M)$ at matched points is frequently nonzero when $C(H)$ is not, foreshadowing the screening-off structure of §4.3.

4.3 Interaction $I(H, M)$ and its anatomy

The central measurement flips a harness decision and a model decision at the same point, alone and jointly. **With budget slack** (max 12 turns; three budget configurations g450/g600/g900), the model action screens off the harness decision *exactly*: $C_{HM} = C_M$ in **57/57** points — the sampled alternative model action re-establishes what the harness destroyed, so the harness flip adds nothing beyond the model flip. **Under budget pressure** (max 4/6/8 turns), screening-off breaks in **8/65** points (slack vs. pressure: $p = 0.005$ point-level hypergeometric; $p = 0.125$ task-clustered sign test over 3 breaking tasks — we report both, and note the point-level test treats points within a task as independent). Breaks recur at the *same* decision points across configurations (`l_log_parser` idx 8 in 3/3 pressure configs; `api_router` idx 1 and `l_vending_machine` idx 11 in 2 each), which is what a real mechanism, rather than noise, predicts.

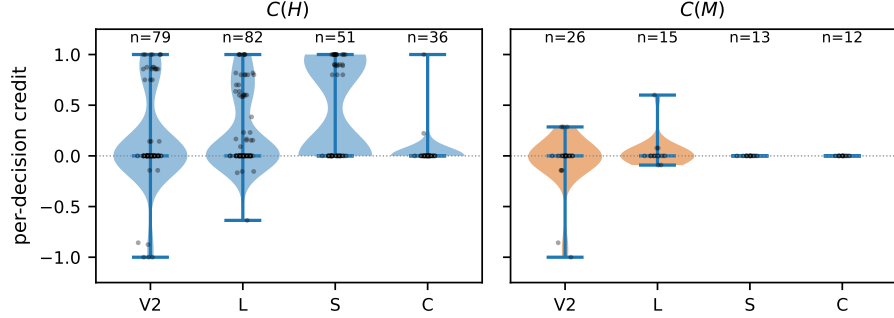


Figure 2: **F2 — Per-decision credit by layer and stratum.** $C(H)$ (248 points) and $C(M)$ (66 points) against an exact zero floor. Strata: V2/L information-destruction, S designed synergy, C controls.

The dependence on pressure intensity is *non-monotonic*: break rates are 0/21 (mt12), 4/22 (mt8), 3/21 (mt6), 1/22 (mt4) — an inverted U. Our pre-registered monotonicity test failed (Cochran–Armitage $p = 0.38$); we report that failure and analyze the cause. The mediator is **reward saturation**: at mt4 the agent fails most tasks outright (R pinned at 0), so no flip can make things worse and interaction is masked. Conditioning on non-saturated points sharpens the contrast (slack 0/12 vs. pressure 6/19, $p = 0.037$) and partially restores the mt4 rate ($0.045 \rightarrow 0.143$), though the ordering $\text{mt8} > \text{mt6} > \text{mt4}$ persists — saturation mediates the window partially, not fully (Figure 3). At the task-cluster level the conditioned contrast, like the raw one, is weak (paired sign-flip permutation over 4 tasks $p = 0.25$; cluster-label permutation $p = 0.18$); the point-level and mechanism-level evidence carry the claim (§7). We call the resulting picture a *competence window*: cross-layer interaction is detectable only where the agent neither saturates success nor failure.

Anatomy of the shield. The sets in play: across the three slack configurations and the first pressure configuration we measured 78 interaction points (57 slack + 21 at mt6); **75/78 show exact screening-off** ($C_{HM} = C_M$), which under the pre-registered sign test against a 50/50 null gives $P = 2.6 \times 10^{-19}$ — the 3 exceptions are the mt6 pressure breaks. Among the screened points, 52 are non-trivially *shielded*: the harness flip alone has a real effect ($C(H) \neq 0$) that the joint flip erases — these are exactly the double-counting cases. Of the 52 shielded points, 24 carry constant-dictionary annotations that make the re-injection check possible (the remaining 28 are unannotated V2 points, which we declare rather than extrapolate). Among the annotated: **23/24 show structural re-injection** — the sampled alternative action a' carries the destroyed information across the cut, a property of the estimand (since a' is sampled from the model conditioned on the un-summarized state); 9 also show the second mechanism and 1 shows neither. The second mechanism is checkable on all 52 shielded

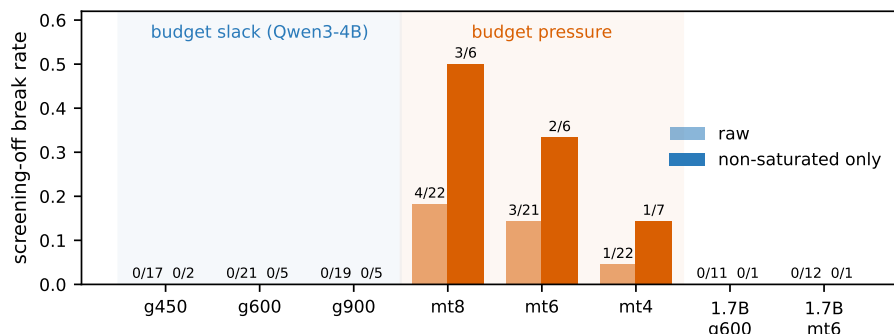


Figure 3: **F3 — Screening-off is regime-dependent.** Break rate of $C_{HM} = C_M$ per configuration, raw and conditioned on non-saturated points. Slack: 0/57 raw (0/12 non-saturated). Pressure: 8/65 raw (6/19 non-saturated, $p = 0.037$). Qwen3-1.7B (purple) replicates slack; its pressure null is explained by saturation (only 1 non-saturated point).

points: in **22/52** the *live downstream harness* re-triggers summarization anyway, equalizing the branches. Screening-off is thus partly a theorem about the estimator and partly an empirical fact about closed-loop harness behavior — and we found genuine non-saturated synergy where it breaks (**1_vending_machine**: $C_H = C_M = -0.09$, $C_{HM} = 0$, so $I = +0.18$: either flip alone hurts, together they cancel).

Consequence for credit assignment. At every shielded point, single-layer harness credit $C(H)$ bills the harness for an effect the model absorbs: interaction-corrected credit $C_{HM} - C_M$ is zero while $C(H)$ is not. In the slack regime this is not an edge case — it is *all* measured points.

4.4 Control tasks and the `c_temp_label` exception

On control tasks, where summaries destroy nothing by construction, the method returns $C(H) = 0$ in 11/12 measured points — the construct behaves. The exception is informative: `c_temp_label` showed $C = +0.22$ at turn 0 despite all task-critical information surviving the summary. Audit of the replay pair shows the effect is *behavioral*, not informational: the summarized context changes the model’s subsequent formatting behavior, which a strict test then penalizes. $C(H)$ measures total causal effect through *any* pathway; the controls certify the informational pathway is clean, and the exception demonstrates the behavioral pathway exists and is detectable.

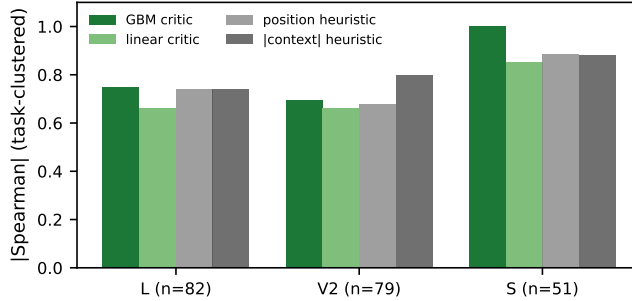


Figure 4: **F4 — Critics vs. trivial heuristics per stratum** (task-clustered $|\text{Spearman}|$). The critic never beats the best heuristic; $C(H)$ is structured by position and context size.

5 The Structure of Harness Credit

Can $C(H)$ be *predicted* from cheap features, so that replay is needed only to calibrate? We trained linear and GBM critics on 248 ground-truth points (features: turn, context size, position, test progress; task-clustered evaluation) against the baselines mandated by [7]: position and context-size heuristics. The honest answer is that **the critic never beats the heuristics** (Figure 4): on the combined L+V2 strata ($n = 161$, task-clustered Spearman) the GBM reaches 0.667 and the linear critic 0.675, against 0.708 for the position heuristic and 0.770 for $|\text{context}|$; pooled over all 248 points the linear critic’s 0.718 still trails the context-size heuristic. Per stratum, L is a tie (GBM 0.747 vs. position 0.741), in V2 the heuristic wins (0.693 vs. 0.796), and S saturates for everything (AUROC 1.0). For detection ($|C| > 0$) the GBM has a modest edge (AUROC 0.846 vs. 0.735 for position and 0.785 for the sign-inverted context-size heuristic) with overlapping CIs.

We report this not as a failed component but as a finding about the signal: **harness credit is mostly a function of context size and position**. That is exactly what the information-destruction mechanism predicts — how much is destroyed, and how late — and it means practitioners can get most of the ranking value of $C(H)$ from features they already log, reserving replay for calibration and for the interaction term, which no cheap feature predicts.

6 Training with Cross-Layer Credit

Does better credit train a better harness? We train the harness `context_policy` (a logistic policy over 5 features: bias, turn fraction, context tokens, test progress, message count) with four arms, **dose-matched in total rollouts** (episodes + credit replays): (1) *outcome-only* policy gradient; (2) *single-layer* $C(H)$ credit — an explicit reproduction of the CHILL-Harness principle in our environment;

(3) *interaction-corrected* credit $C_{HM} - C_M$; (4) a *zero-credit* masking control (credit $\equiv 0$, no replays — which under our greedy tie-break at $\theta = 0$ is de facto an untrained keep-always policy; we state this explicitly). The pre-registered success criterion (GATE 3) required arm 3 to beat both arm 2 and arm 4. Calibration established the target is learnable in principle: on held-out tasks a simple context-size threshold policy (thr600) achieves $R_{\text{eff}} = 0.398$, dominating keep-always (0.237) and summarize-always (-0.464); thr600 is expressible by the policy class.

Act 1: seed-dependent collapse. With raw features and $\text{lr} = 0.5$, every trained arm collapsed into one of the two fixed policies, with the attractor determined by seed (outcome: k/k/s; $C(H)$: s/s/k; $C_{HM} - C_M$: k/s/s). GATE 3 failed and we report it. Two hypotheses remained: the credit estimator is broken, or the optimization is. A pre-registered audit recomputed 60 logged credits by fresh two-greedy-replay: **58/60 sign agreement, zero sign flips toward harmful** (mean discrepancy -0.0015). The estimator is exonerated.

Act 2: stable optimization, no discovery. With the pre-registered fix (feature centering, $\text{lr} = 0.1$, gradient clipping), training is sane: gradients bounded, learned weights point the right way (negative bias, positive weight on context tokens, implicit decision thresholds at 2.5k–5.8k tokens). Yet **no arm in any seed discovers the threshold policy**: all non-collapsed runs converge to keep-always (held-out $R_{\text{eff}} = 0.237$ vs. thr600’s 0.398; Figure 5). Worse, credit-driven arms retain collapse risk the outcome-only arm avoids ($C(H)$: 1/3 seeds; $C_{HM} - C_M$: 1/3; outcome-only and zero: 0/3).

Diagnosis. Dose-matching in total rollouts means credit arms pay for replays with episodes: across 3 seeds the outcome arm saw 1,789 training episodes, $C(H)$ 786, $C_{HM} - C_M$ 319 (zero: 1,910). One might suspect the credit arms simply saw too little data. The state-visitation record rules this reading out and localizes the failure more precisely than “exploration”: across *all* arms and seeds, **0 of 13,498 context_policy** decisions occurred at context sizes $\geq 2,500$ tokens. The zero arm — de facto keep-always, the policy that maximizes context growth — caps at 1,532 tokens over 5,270 decisions and 1,910 episodes. The region where the threshold policy pays is therefore not under-explored on the training tasks; it is *unreachable under any context policy*, because the 20 training tasks (an alphabetical split, fixed before training and before we knew this gap existed) never grow contexts that large, while held-out tasks do (e.g. `1_vending_machine` reaches 5,933 prompt tokens under keep-always, $R_{\text{eff}} = -1.12$). No exploration bonus and no amount of additional data from the same task distribution would help. **The binding constraint is state coverage of the training task distribution, not credit quality.** This sharpens the design lesson: a cross-layer training benchmark must verify that the state support of its training tasks overlaps the states where candidate policies actually differ — otherwise credit fidelity is moot, and the failure will masquerade as an algorithmic one.

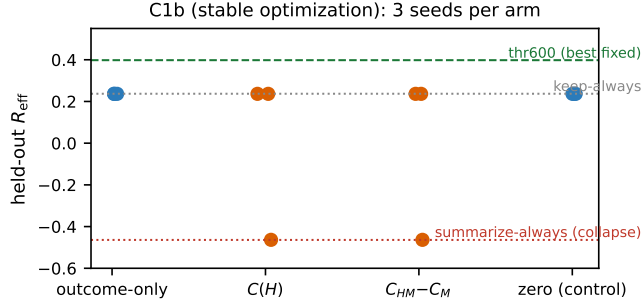


Figure 5: **F5 — Training outcome (C1b, stable optimization).** Held-out R_{eff} , 3 seeds per arm. No arm reaches the expressible thr600 policy; credit-driven arms (orange) carry collapse risk that outcome-only avoids.

7 Threats to Validity and Discussion

Synthetic tasks and external validity. Our tasks are designed, not sampled from a public benchmark, and this is a real limitation — but a motivated one. The measurement requires (i) control over the *position* of destroyable information, to separate informational from behavioral effects; (ii) per-stratum calibrated recoverability, which underlies the control tasks; and (iii) deterministic fractional reward, which underlies the exact floor. No public benchmark provides these jointly. The exact missing experiment is a multi-turn MBPP+ adaptation (~ 100 tasks, 1–2 GPU-days); we declare it rather than claim it. We position this as a methods paper: the transferable artifact is the recorder/replay/forced-queue infrastructure, the APC finding, and the measurement protocol.

One model powers the pressure regime. The floor and slack regime replicate on Qwen3-1.7B; the pressure-regime break rate is established on Qwen3-4B only. The 1.7B null (0/12) is itself explained by the mediator we identified: 11/12 of its pressure points are reward-saturated, leaving one informative point — zero power (under the 4B break rate, $P(0 \text{ breaks}) \approx 0.21$). This is the competence window operating as predicted, but a second *powered* model is needed to call the pressure regime general.

Clustered inference. Points within a task are not independent. At the task level the pressure-vs-slack contrast is 3 tasks vs. 0 (sign test $p = 0.125$); at the point level $p = 0.005$; the saturation-conditioned contrast behaves the same way (point-level $p = 0.037$; task-clustered permutation $p = 0.18\text{--}0.25$). We report all levels and lean on the screening-off sign test (75/78, $P = 2.6 \times 10^{-19}$) and cross-configuration recurrence of break points as the stronger evidence.

Estimator coupling. $C(M)$ and I share the sampled alternative action a' : they are not independent evidence, and the structural component of screening-off (§4.3) is partly a property of the estimand — a' sampled from the unsummarized state can re-inject information. We quantified this (23/24 annotated shielded points; 28 unannotated points declared) rather than hiding it; an estimand with a' sampled from the summarized state would measure a different, also legitimate, quantity.

Saturation is endogenous. Reward resolution is set by test count, so saturation — our mediator — is partly a design artifact. The competence-window analysis conditions on it explicitly; the inverted-U at mt4 is only partially explained (conditioned rate $0.045 \rightarrow 0.143$; the ordering $\text{mt8} > \text{mt6} > \text{mt4}$ persists), so the window remains a partially supported hypothesis, discussed but not claimed as a contribution.

Serving premises. The exact floor holds under fixed config, sequential requests, and APC off. It is a measured property of this stack, not a theorem; any deviation (batching, caching, config drift) voids it and must be re-measured.

8 Conclusion

We built the first apparatus that measures, on the same trajectory, the causal credit of harness decisions, model decisions, and their interaction, against an exactly-zero measured noise floor. The measurement yields a crisp, regime-dependent picture: with budget slack the model screens off the harness completely (0/57), so single-layer harness credit double-counts; under budget pressure the shield breaks at recurring points (8/65), inside a competence window shaped by reward saturation. Better credit did not, however, train a better harness: a two-stage, pre-registered negative — with the estimator exonerated by audit — locates the binding constraint in training-distribution state coverage, not attribution. We offer the infrastructure, the regime-dependence finding, the APC determinism warning, and a pre-registration ledger of 14 predictions (including 3 failures) as the contributions, and the training negative as the design brief for the benchmark this field still needs.

A Pre-Registration Ledger

All confirmatory analyses were pre-registered in the project’s execution plan before the corresponding data were collected. Table 2 lists all 14 pre-registrations with their outcomes, including the three that failed. Failed predictions are reported in the main text where they occur.

#	Pre-registered prediction / rule	Outcome	Status
1	Noise floor per config = max $ \Delta R $ over nulls	0.0 in 7+2 configs, 2 models	held
2	Phase-A yield / stopping rule	followed as written	held
3	Critic evaluation statistic (task-clustered Spearman)	mixed result reported as-is	held
4	Pre/post feature split for critic	followed	held
5	Structured a' = distinct estimand, reported separately	followed	held
6	C1 arms dose-matched in total rollouts	followed	held
7	Credit directions never aggregated	followed	held
8	Saturation analyses outside confirmatory set	followed	held
9	GATE 3: arm 3 must beat arms 2 and 4	not met	failed
10	Screening-off mechanism sign test	75/78, $P = 2.6 \times 10^{-19}$	held
11	Estimator audit: sign agreement on 60 recomputed credits	58/60, 0 harmful flips	held
12	C1b primary outcome: held-out $R_{\text{eff}} > 0.30$, $\geq 2/3$ seeds	no arm reached it	failed
13	Dose-response: monotonic break rate in pressure	inverted U ($p = 0.38$)	failed
14	D2c replication: floor, slack, pressure on Qwen3-1.7B	floor+slack replicate; pressure inconclusive	partial

Table 2: Pre-registration ledger. Failures are results, not omissions: #9 and #12 constitute the two-stage training negative (§6); #13 led to the competence-window analysis (§4.3).

References

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